A07616000 - Richtpreisangebot

„Automatic production line for wire processing“

DRÄXLMAIER

08.10.2024

DRÄXLMAIER Group
DIS Dräxlmaier Industrial Solutions Group
84137 Vilsbiburg

Modulare Automation
und Prüftechnik

Gluth SYSTEM TECHNIK

Ansprechpartner

DRÄXLMAIER Group

Renate Schandl
Einkauf
+49 8741 47 – 3574
+49 152 2800 4115
Renate.Schandl@draexlmaier.com

Juhasz Levente
Technik
+40 723 100 146
Levente-istvan.juhasz@draexlmaier.com

Roland Györi
Technik
+40 732 723 879
Levente-istvan.juhasz@draexlmaier.com

Gluth Systemtechnik

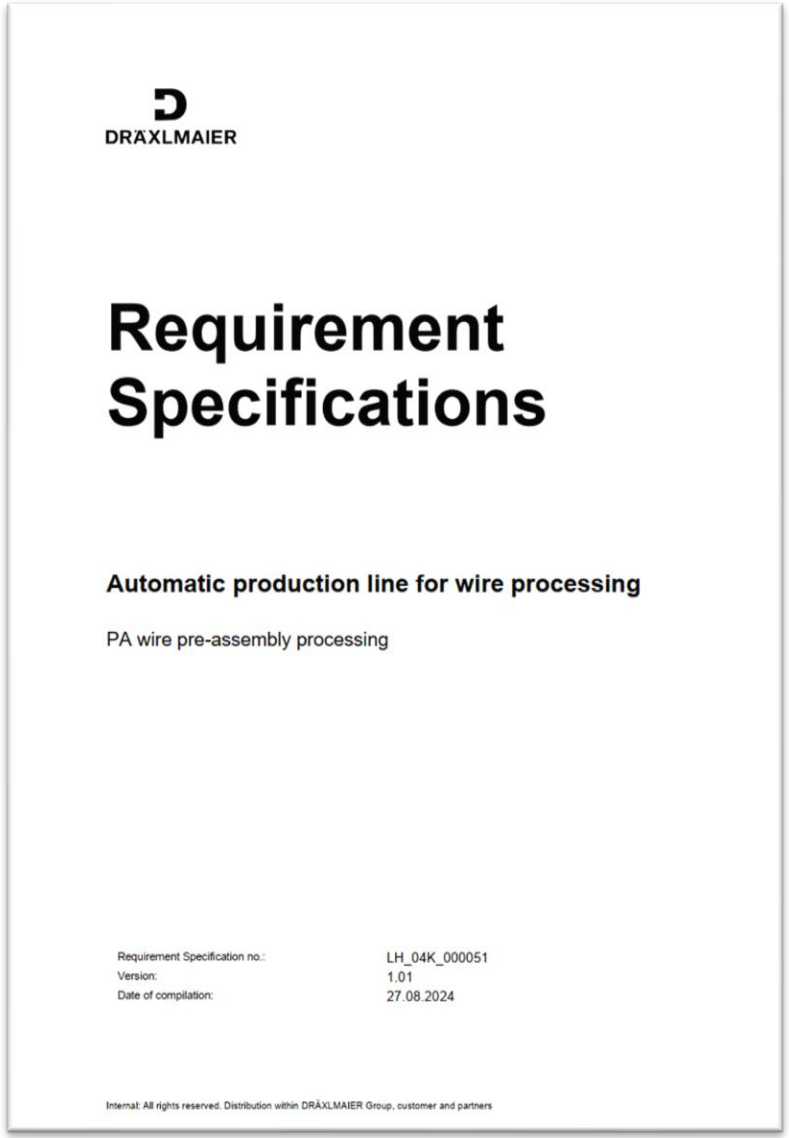
Rudolf Schano
Technischer Vertrieb
+49 9424 544 953
+49 151 4026 3053
Schano.Rudolf@gluth.eu

Grundlagen

Lastenheft **„Automatic production line for wire processing“**
Version 1.01 vom 27.08.24

Taktzeit: < 25 sec.

Standort: Rumänien, Satu Mare



DRÄXLMAIER

Requirement Specifications

Automatic production line for wire processing

PA wire pre-assembly processing

Requirement Specification no.:	LH_04K_000051
Version:	1.01
Date of compilation:	27.08.2024

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Aufgabenbeschreibung

9.2 Product drawing

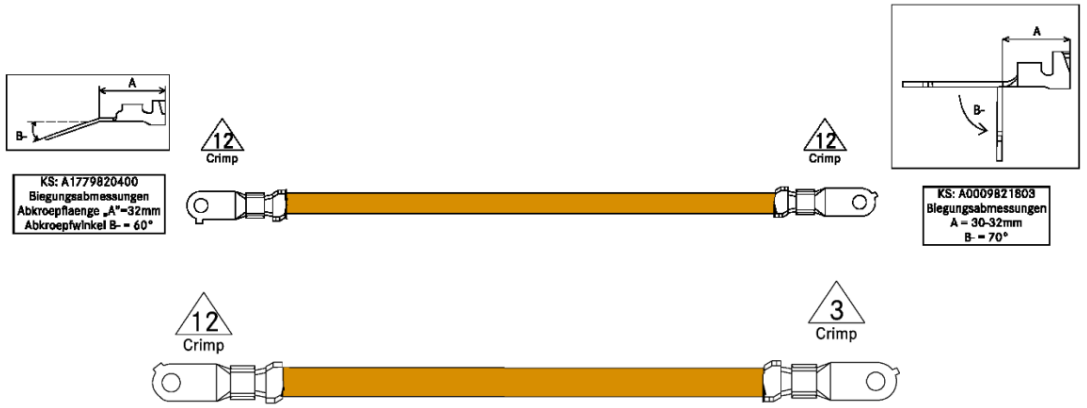
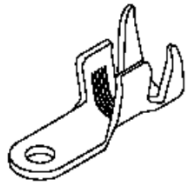
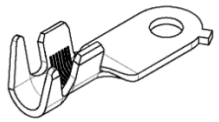


Figure 1: Assembled product

9.3 Assembly components

Designation	Amount	Material number
Cable (Low Voltage singlecore cable 16mm²/25mm² /35mm² FLRY-B;FLR2G;) 	1 pc	L0263CXX (16mm²) L3363066 (16mm²) L3366063 (25mm²) L3369048 (35mm²)
TB Cable lug, special shape (terminal)  	1 pc	A4813709 A4813711 A4813718 A4813719 A4813725 A4816710 A4816727 A4816743 A4817942 A4817952

Aufgabenbeschreibung

10 Process description

Process description

ST01 - Insertion station:
1. Manual scanning of the identification label placed on the cable into the DRX MES.
   - By scanning, the equipment receive the input data from DRX MES and load automatically the right processing program for the product variant
2. Manual insertion and positioning of the cable into the insertion unit (with possibility of buffer at least 20pcs).
   - In case that is processed the second end of cable must ensure the right orientation between the cable lugs on the two ends.
   - Insertion may only be possible in the correct position and the correct orientation (Poka Yoke)
3. Automatic fixing of cable by reaching the right position and orientation
4. Automatic checking of cable diameter correctness.
5. Automatic transfer of cable to the next station.

ST02 – Stripping station:
1. Automatic insert of cable end into the stripping unit.
2. Automatic stripping cut of insulation at the defined position.
3. Automatic removing of insulation part.
4. Automatic transfer of cable to the next station.



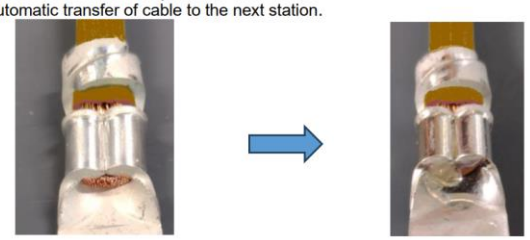
ST03 – Crimping station:
1. Automatic feeding of cable lug on the crimping position
2. Automatic insert of cable end into the crimping unit
3. Automatic crimping/pressing of cable lug on cable
4. Automatic transfer of cable to the next station.



ST04 – Bending station:
1. Automatic insert of cable end with crimped cable lug into the bending unit
   - If no bending is requested by the order, the product is transferred automatically to the next station.
2. Automatic bending of cable lug on the predefined position “A” to the defined angle “B” in the defined direction “+” or “-“.
3. Automatic transfer of cable to the next station.

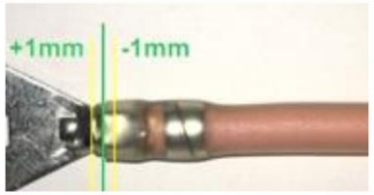


ST05 – Tinning station (soldering):
1. Automatic insert of cable end with crimped/bent cable lug into the tinning unit
2. Automatic tinning/soldering of the wires strands outcoming from crimped cable lug
3. Automatic cool down of product
4. Automatic transfer of cable to the next station.



ST06 – Quality inspection station:
1. Automatic insert of cable end with crimped/bent/tinned cable lug into the inspection unit
2. Automatic inspection of visual aspects of product
   - 360° vision system have to be implemented
3. Automatic transfer of cable to the next station.

☒ Option: ST07 – Shrinking station:
1. Automatic positioning of shrink tube on the cable lug
   - If no shrinking is requested by the order, the product is transferred automatically to the next station.
2. Automatic insert of cable end with crimped/bent/tinned cable lug into the shrinking unit
3. Automatic contraction of the shrink tube
4. Automatic cool down of product
5. Automatic transfer of cable to the next station.



ST08 – Output station:
1. Automatic release of OK product and place on the defined position (collecting recipient)
2. Automatic separation of NOK product in the defined position (NOK box)

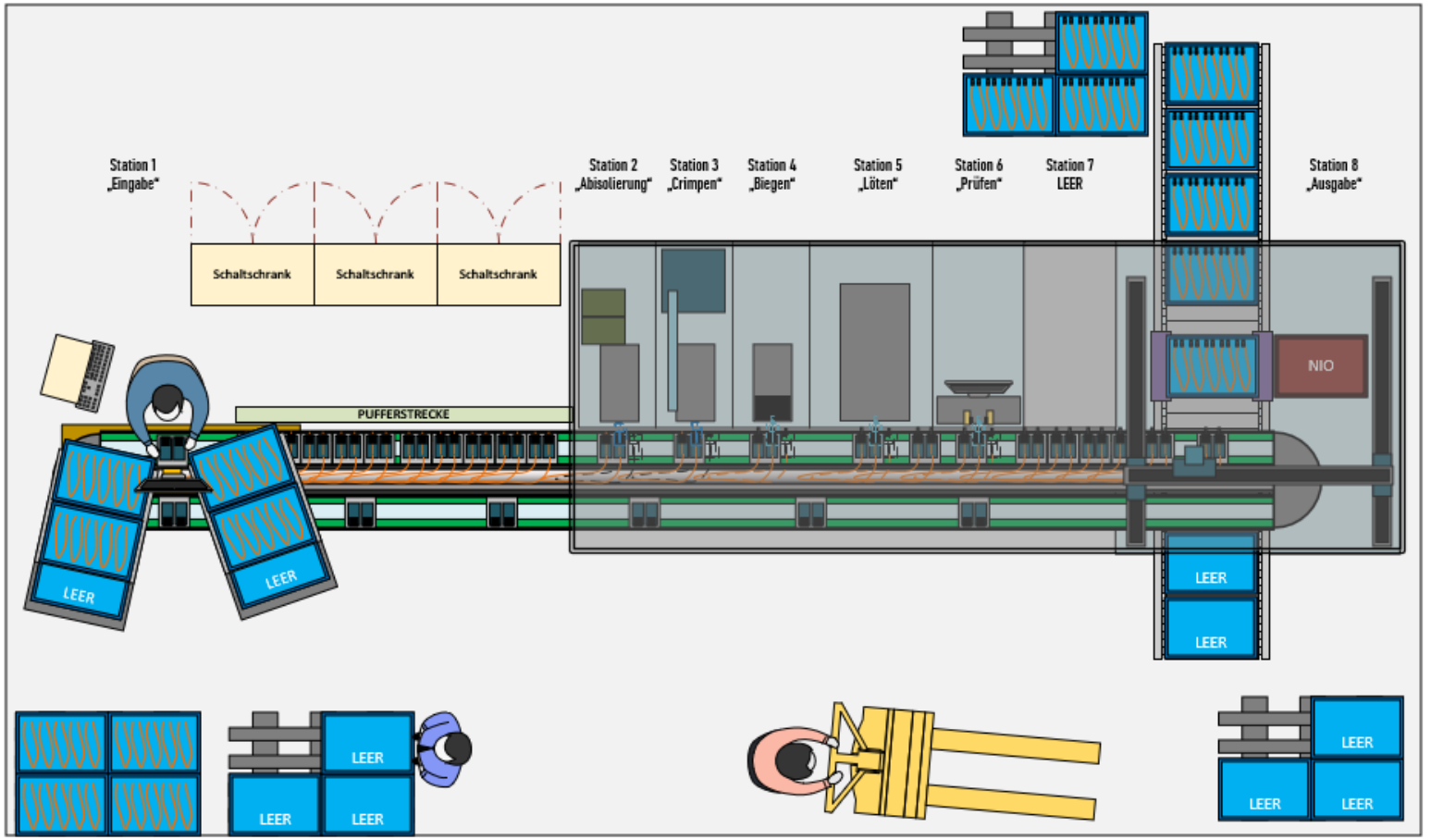
Variante 1

Variante 1:
„Vollautomatisiert“
Vorgaben laut Lastenheft

Eingabe Station 1: Manuell

Station 2 – 8: Automatisch





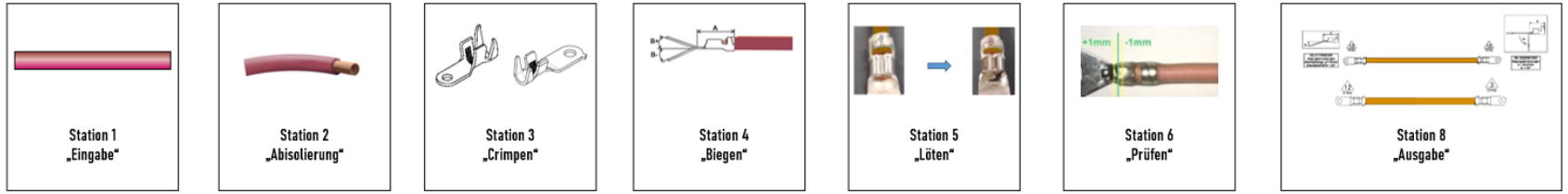
Variante 2

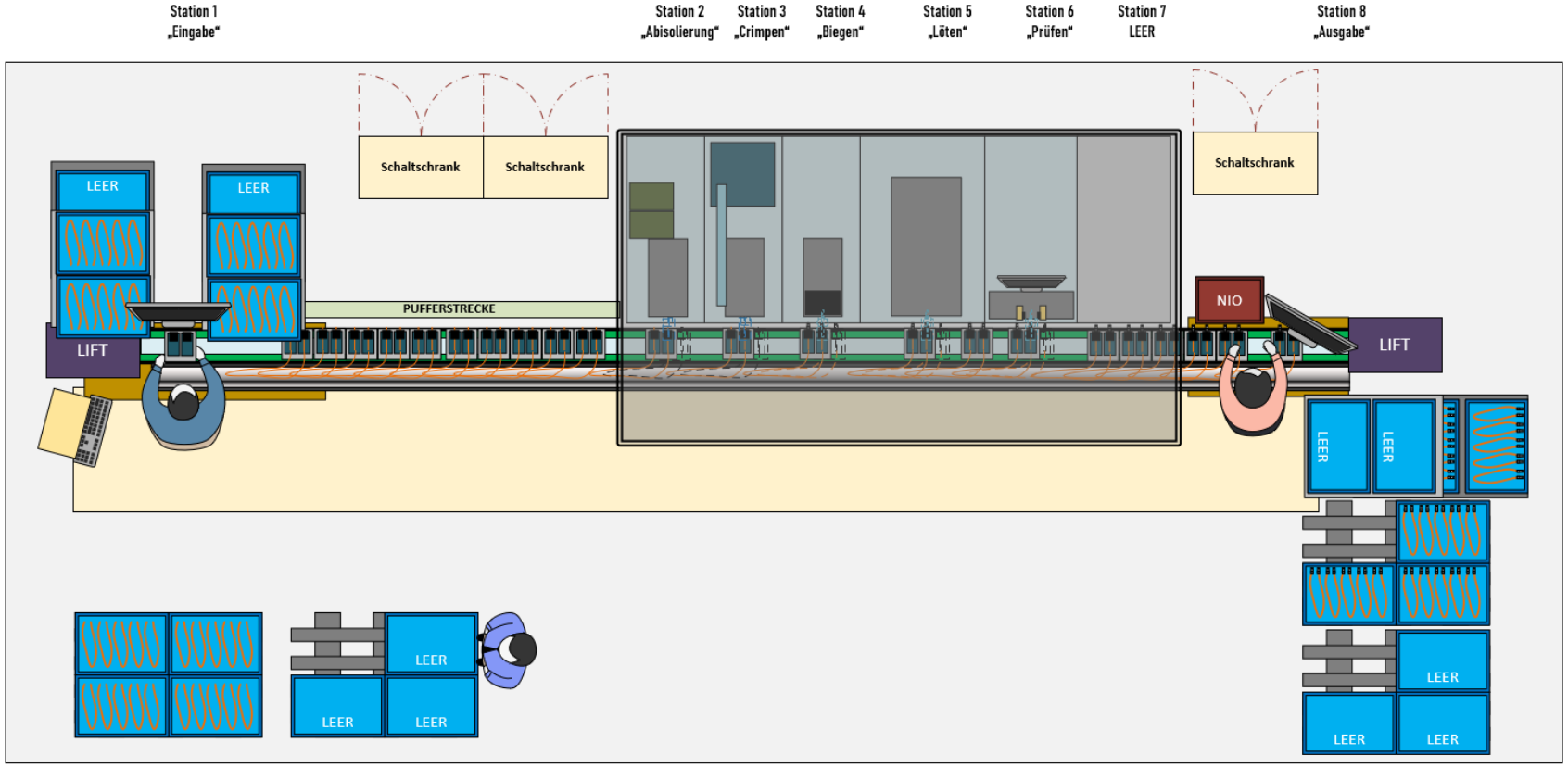
Variante 2:
„Teilautomatisiert“

Eingabe Station 1: Manuell

Station 2 – 6: Automatisch

Ausgabe Station 8: Manuell





Variante 3

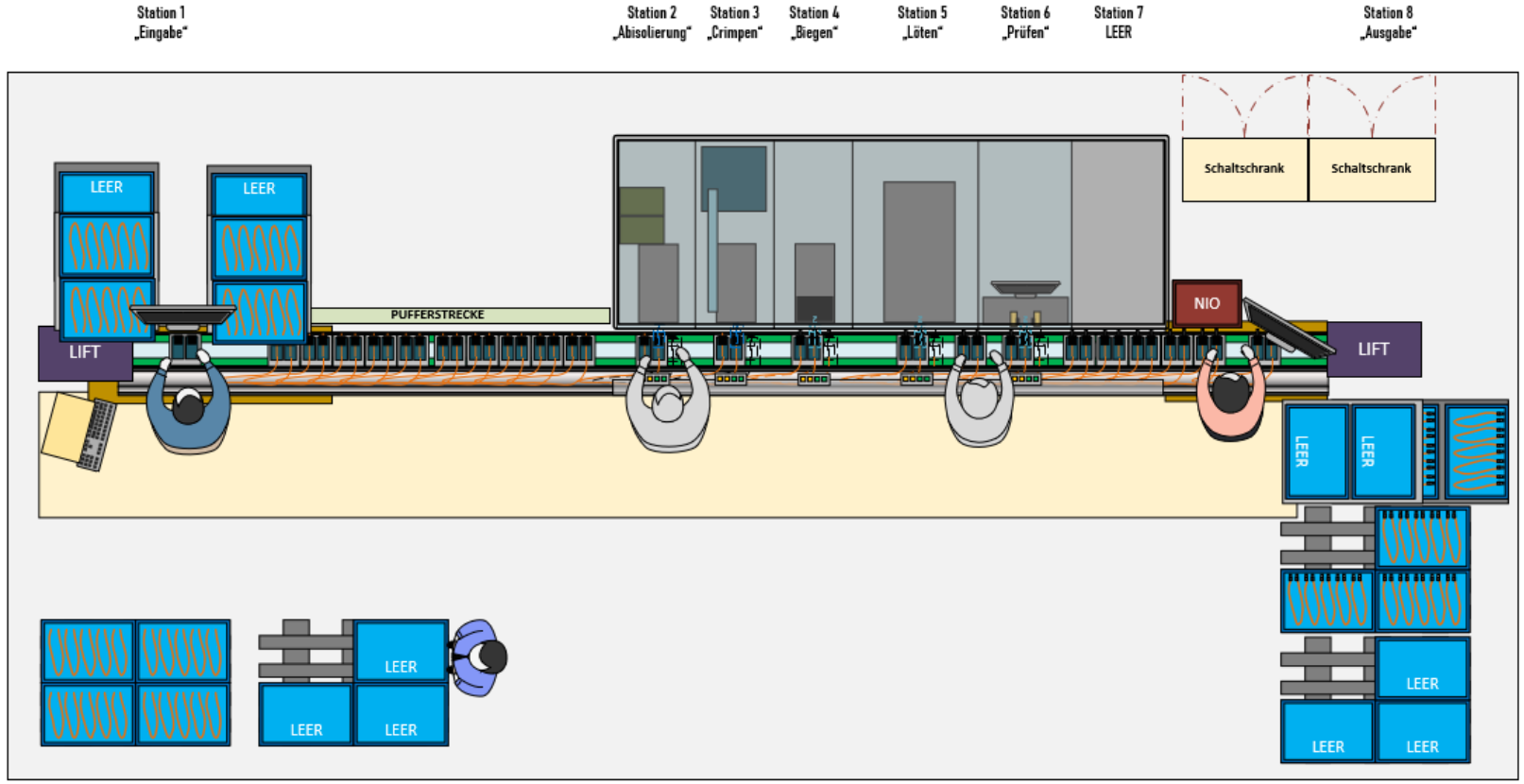
Variante 3:
„Teilautomatisiert mit Handzuführungen“

Eingabe Station 1: Manuell

Station 2 – 6: Zuführung Manuell mit Automatischen Prozessen

Ausgabe Station 8: Manuell





Variante 4

Variante 4:
„Teilautomatisiert mit kpl. manuellen Bauteilhandling“

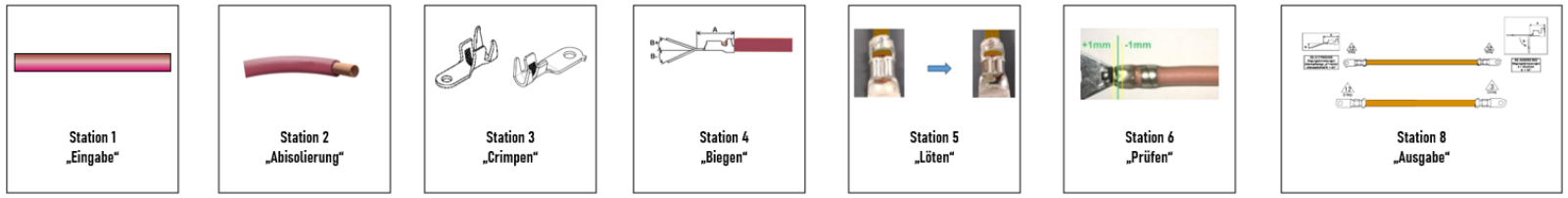
Eingabe Station 1: Manuell

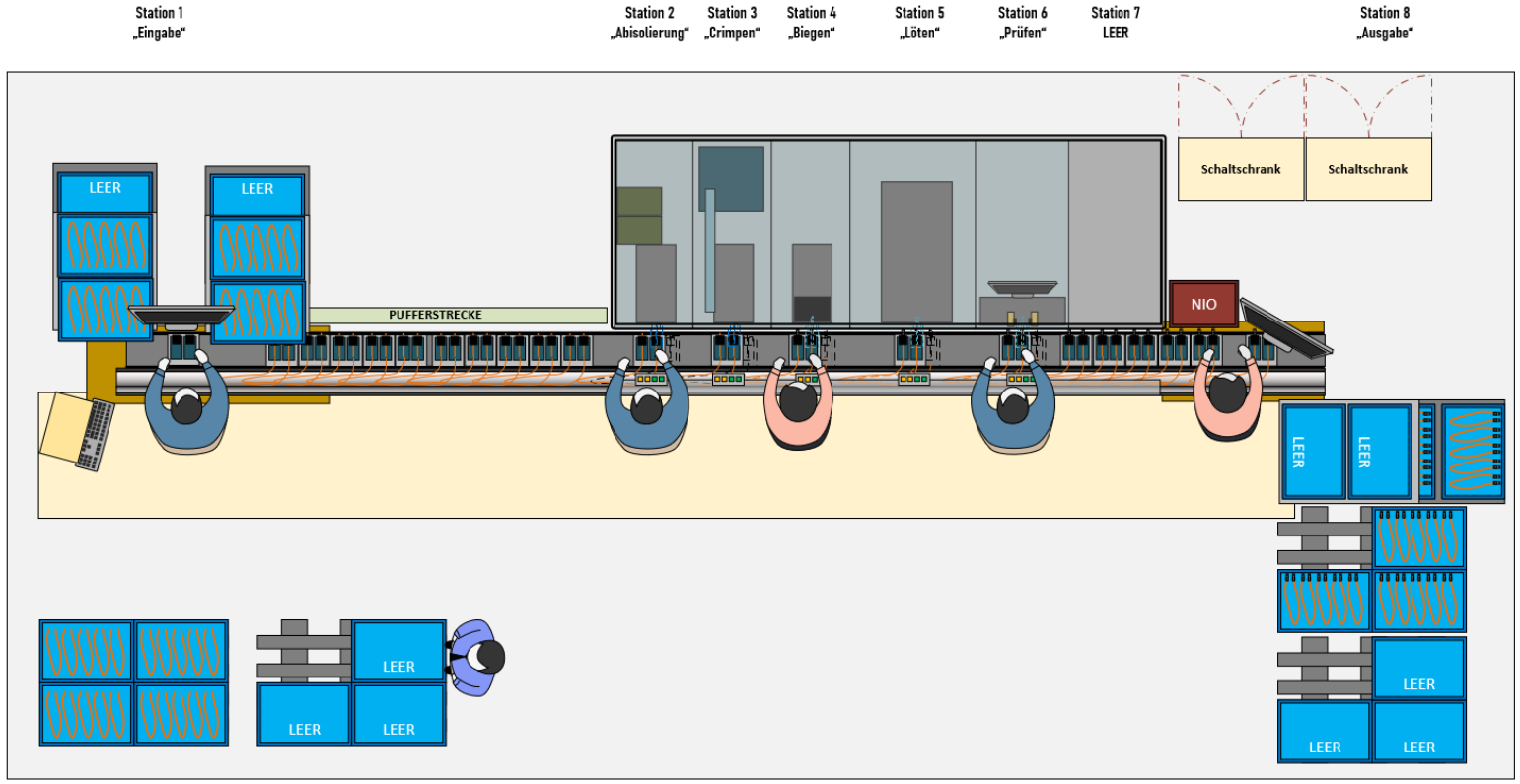
Station 2 – 6: Zuführung Manuell mit Automatischen Prozessen

Ausgabe Station 8: Manuell

Manuelles Verschieben Werkstückträger

Manuelle Rückführung leerer Werkstückträger





Richtpreise

Richtpreis je nach Ausbauvariante

Variante 1 „Vollautomatisiert“: ca. 1.250.000 €
Variante 2 „Teilautomatisiert“: ca. 1.130.000 €
Variante 3 „Teilautomatisiert mit Handzuführung“: ca. 1.025.000 €
Variante 4 „Teilautomatisiert mit kpl. manuellen Bauteilhandling: ca. 895.000 €

Der Richtpreis beinhaltet:

- *Erstellen der Mechanischen Konstruktion*
- *Erstellen der Hardwareplanung und der Software*
- *Beschaffung der Kauf- und Fertigungsteile (Mechanisch und Elektrisch)*
- *Montage mit elektrischer, pneumatischer Installation bei uns im Hause*
- *Endmontage und Inbetriebnahme sowie Vorabnahme der Anlage bei uns im Hause*
- *Demontage und Verladen auf LKW*
- *Wiederaufbau und Inbetriebnahme in Ihrem Werk*
- *Schulung/Einweisung Ihrer Mitarbeiter*
- *Erstellung der Dokumentationsunterlagen*
- *Projektmanagement*

Hinweis:
Aktuell noch geschätzte Preise, da die Abklärungen mit den Prozessen Abisolieren, Crimpen, Biegen, Löten und Prüfen noch nicht abgeschlossen ist.
Eine Technische Durchsprache der einzelnen Prozesse fand noch nicht statt.



Danke
für die Aufmerksamkeit

Gluth SYSTEM TECHNIK

Gluth Systemtechnik GmbH
Steinweg 66 | 94315 Straubing
+49 9421 544-0 | info@gluth.de | www.gluth.de